Project 2 Report

Introduction

For this project, our chosen object is a welded metal bed frame. The frame functions as a structural support system that transfers loads from the mattress and occupants to the floor through rails, cross members, and vertical legs. We modelled ¼ of this bed in CAD for simplicity, shown below.



The motivation for analyzing this object comes from its simplicity, everyday relevance, and strong potential for optimization. Bed frames are designed to support significant loads over long periods, yet many commercial frames are overbuilt to ensure safety across a wide range of use cases. This creates an opportunity to reduce material usage while maintaining structural integrity

The primary optimization objective for this study is to minimize the mass of the frame. However, this must be achieved while satisfying several structural constraints. Specifically, the maximum stress in the frame must remain below the yield strength of the material to prevent permanent deformation. Additionally, deflection must be limited to maintain user comfort and structural stiffness, and the overall stability of joints and members must be preserved. These constraints represent the classic engineering tradeoff between lightweight design and structural performance.

Background

We compared our results to studies previously conducted. One of these studies was looking at a wooden bed frame but still provided good context for our study. This was an FEA study.

Matwiej, Ł., Wiaderek, K., Jarecki, W., Orlikowski, D., & Wieruszewski, M. (2025). *Modelling upholstered furniture frames using the finite element method*. **Applied Sciences, 15**(2), 926. https://doi.org/10.3390/app15020926

The stress results from their study show a similar profile to our study and provided context for which part is likely to be harder to reduce our mass for without failing to meet our constraints. This result from the reference study can be seen below.



The second reference study we looked at helped us define the load was a study analyzing the strength of bed frames. This study used human body weight as the design load for bed structures and simulations.

Réh, R., Krišťák, L., Igaz, R., & Hitka, M. (2019). *Analysis to improve the strength of beds due to the excess weight of users*. **Sustainability, 11**(3), 624. https://doi.org/10.3390/su11030624

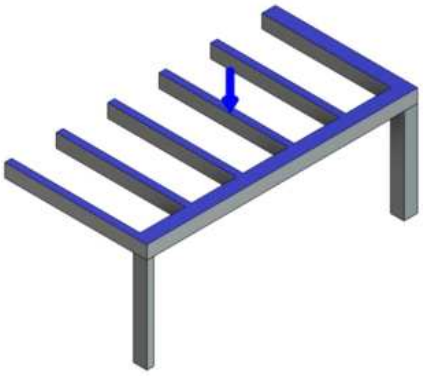
The study used a distributed weight of 243lb which we increased to 300lb to increase the factor of safety.

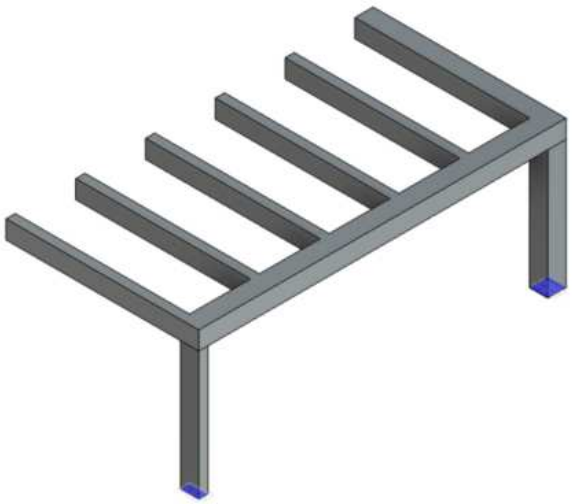
Method

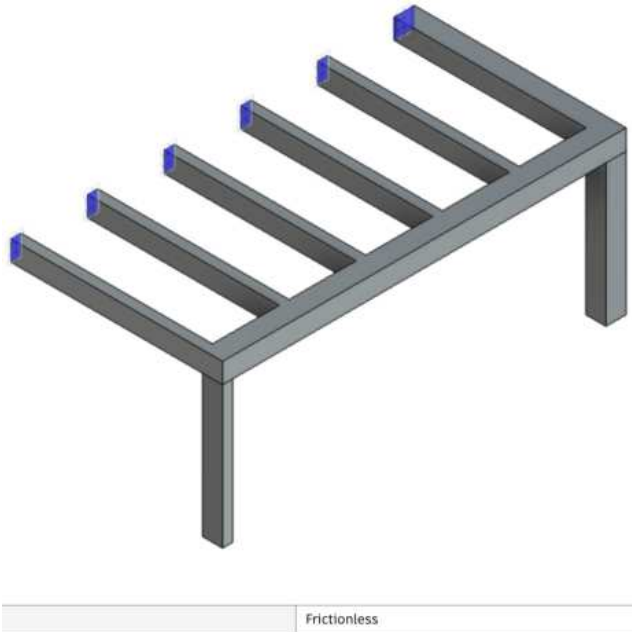
The material selected for this analysis is ASTM A36 structural steel, which is commonly used in welded consumer products due to its low cost, good weldability, and reliable mechanical properties. A36 steel has an approximate yield strength of 250 MPa and an elastic modulus of about 200 GPa, making it a reasonable approximation for the mild steel typically used in commercial bed frames.

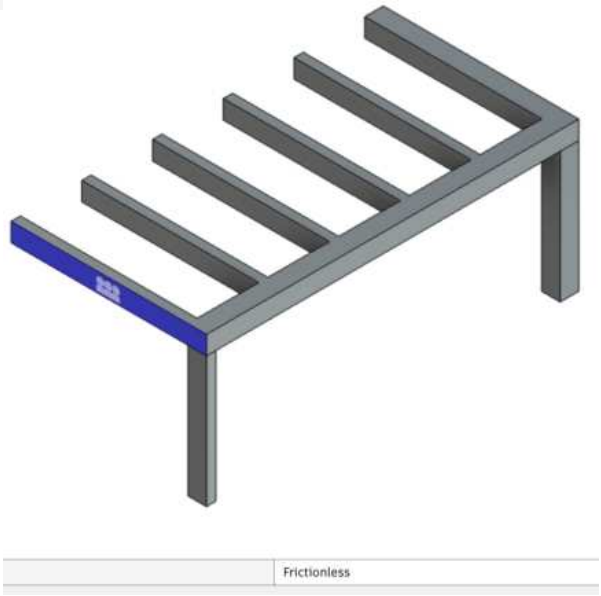
In terms of loading and boundary conditions, the bed frame experiences a distributed vertical load that represents the combined weight of the mattress and occupants. In the simulation, this is modeled as a uniform distributed force applied along the top rails of the frame. The feet of the bed are constrained at the floor using fixed or sliding boundary conditions, preventing vertical motion while approximating realistic contact with the ground. This setup reflects how loads are transferred in real-world use, where weight is spread across the mattress and carried through the frame to the supporting legs.

These can be seen below:





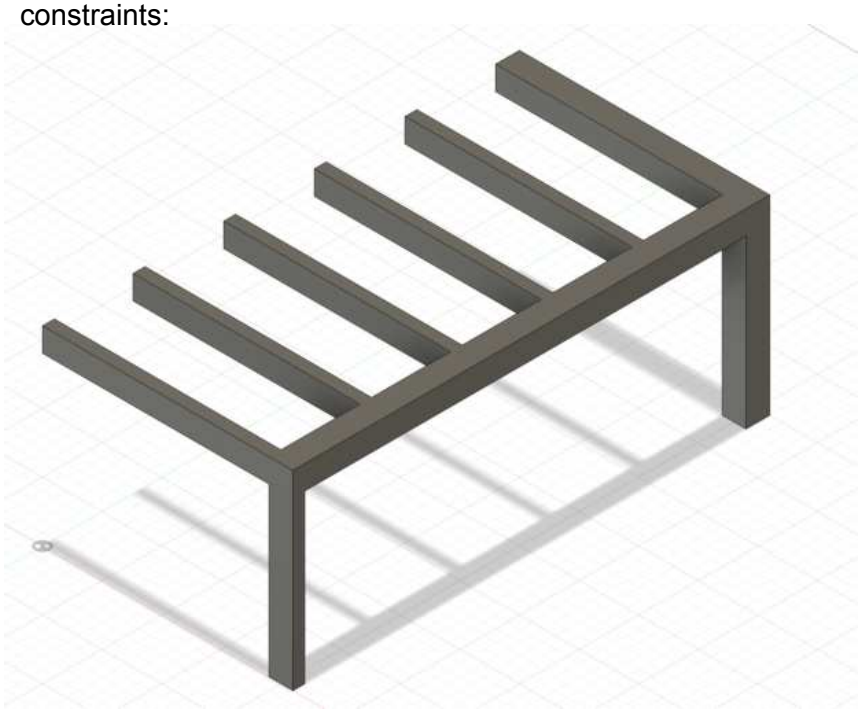


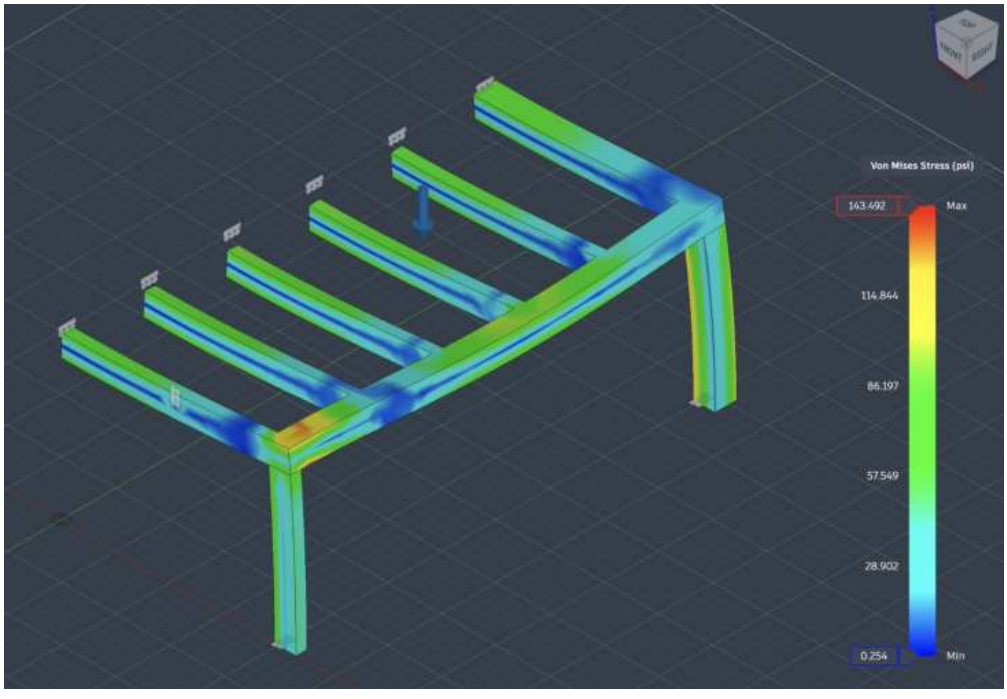


The objective was to decrease mass by at least 20% whilst having a maximum stress less than the yield stress.

The software used was Fusion 360 in which we used Shape Optimization. The objective to minimize was the mass of the object, with the constraint that the stress must not exceed the yield stress. An initial iteration with no preserved geometry was run. This iteration led to the main support legs being completely removed, which is unrealistic. Thus a second iteration was run with preserved geometry around the legs, focusing the shape optimization on the top portion of the frame directly supporting the mattress.

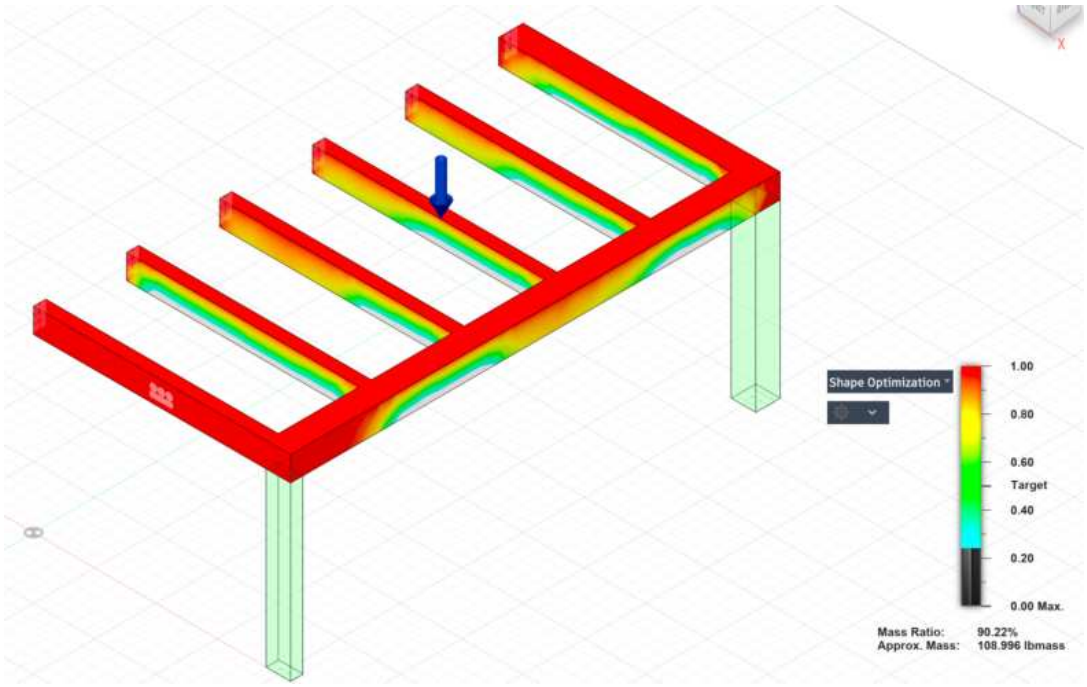
Below is a visual of our initial model and the Von Mises stress under our stated loads and constraints:



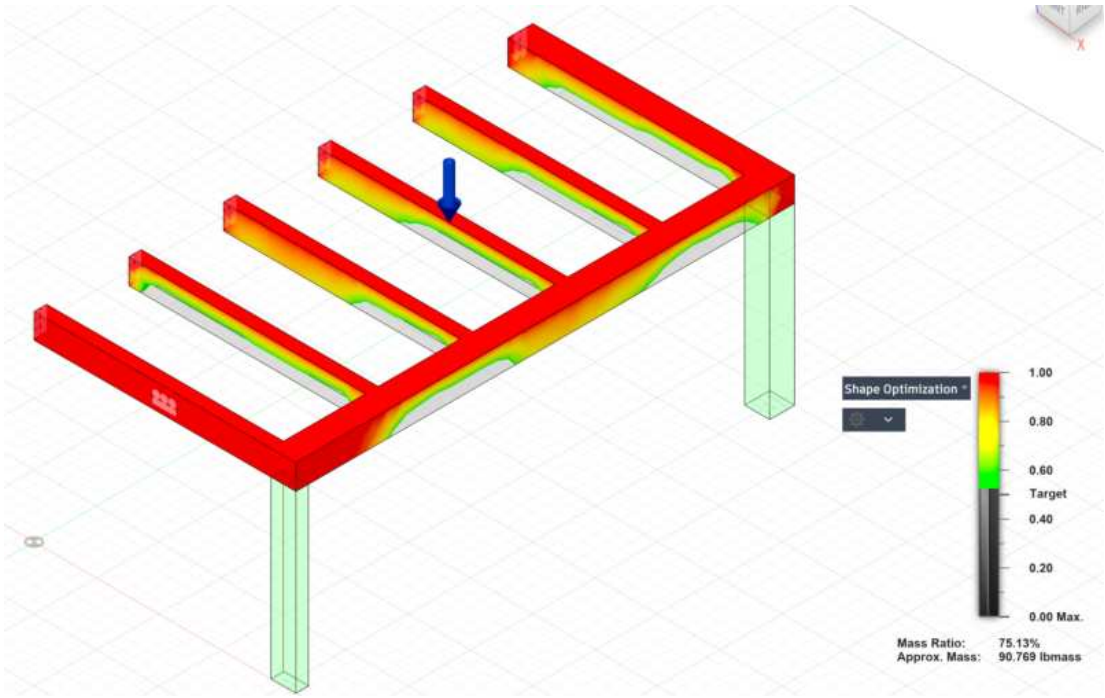


Results

We tried two different percent mass thresholds in our optimization, initially 90%:

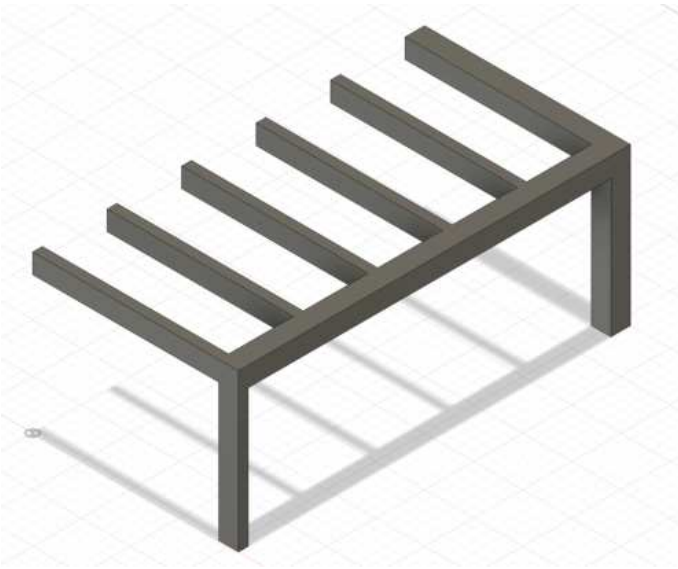


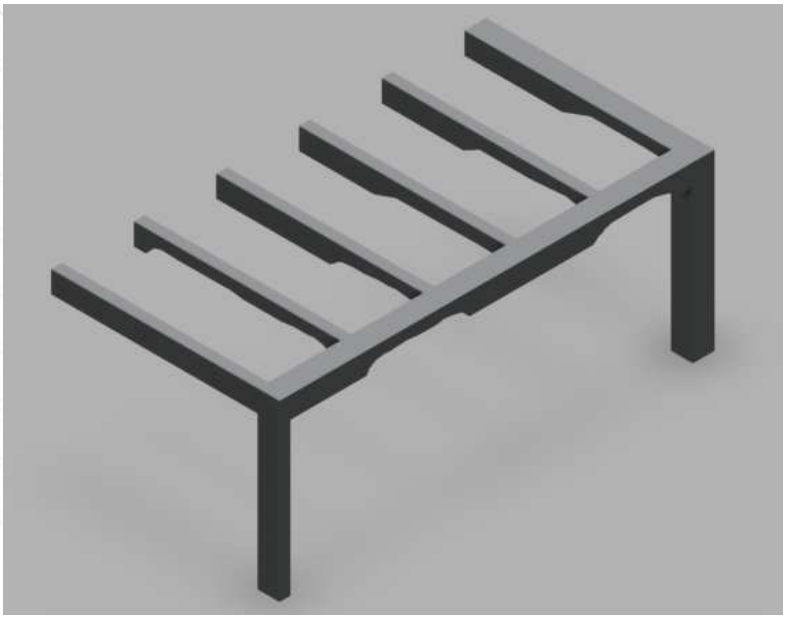
Then we saw a high factor of safety so further decreased the mass threshold to 75%, this is what we chose as our final design:



Comparison

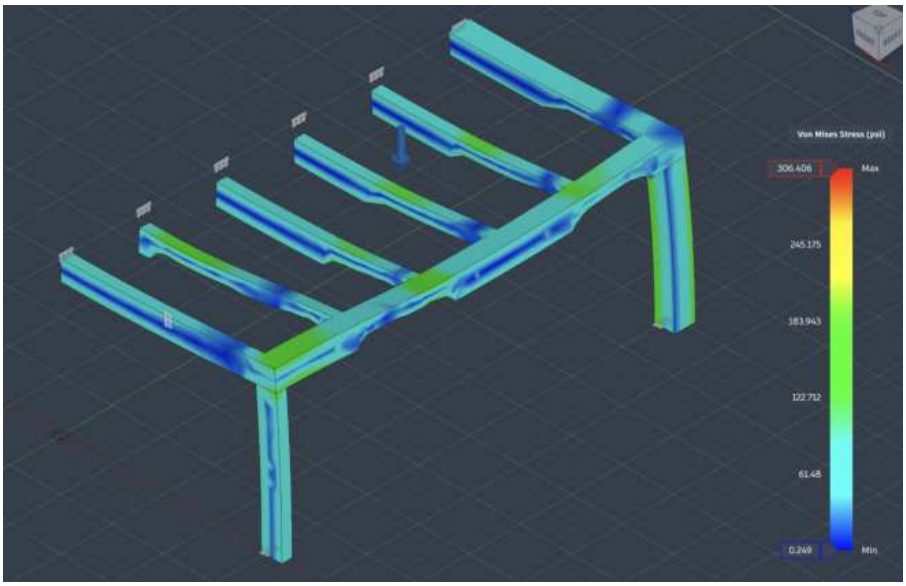
Below is both our initial and final design side by side to compare:





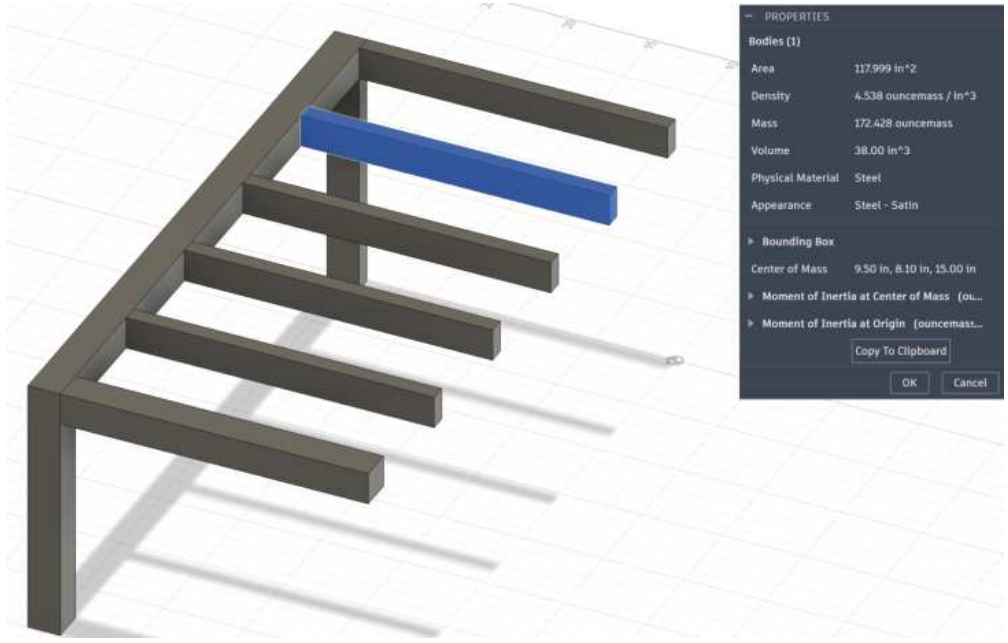
As we can see most of the mass was removed at the strongest parts of the beams, which had lower stress concentrations. The main quantitative difference between these two is the final design having 75% the mass of the initial design.

Additionally, if we look at the Von Mises stress of the final design below. We can see that the max stress of the final design is 306 psi whereas the initial design had a max stress of 143 psi. However both of these are well under the yield stress.

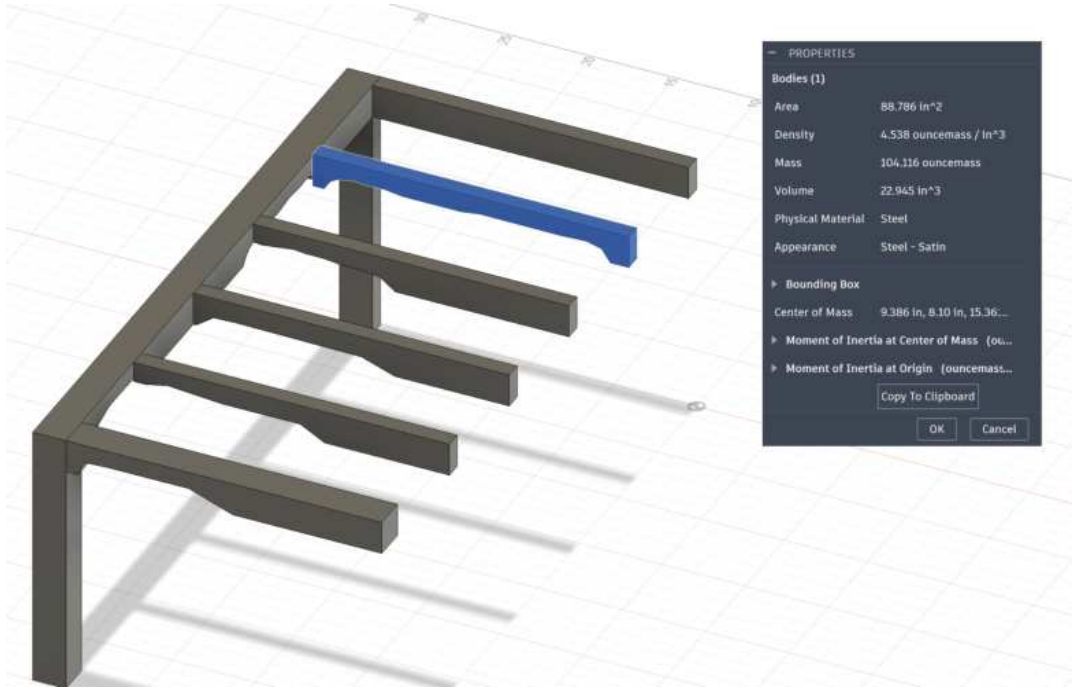


We also looked to compare a specific area of interest between the two designs. We chose the bar which had the most mass removed after the shape optimization process. This change in the two designs can be seen highlighted below:

Initial Design:



Final Design:



We can see from these two images that the mass of this specific bar was 172 oz in the initial design and 104 oz in the final design, which is a 39.5% decrease in mass of this specific area.

Conclusion

The objective of this project was to reduce the mass of a welded metal bed frame while ensuring stresses remained below the yield strength of ASTM A36 steel. Using shape optimization in Fusion 360, the final design achieved a 25% reduction in mass, meeting the optimization goal while maintaining structural safety. Although the maximum Von Mises stress increased from 143 psi in the initial design to 306 psi in the final design, both values remain far below the material yield strength, indicating the frame can safely support the applied loads and constraints.

The main performance difference between the initial and optimized designs came from removing material in low-stress regions of the beams while preserving geometry near the legs and joints where stresses were higher. A key example was one cross member where mass decreased from 172 oz to 104 oz, a 39.5% reduction while still meeting the stress constraint.

One challenge encountered was that the initial optimization removed the support legs entirely, producing an unrealistic design. This was addressed by preserving geometry around the legs. Future work could improve the model by applying more realistic load distributions and validating the results with physical testing.